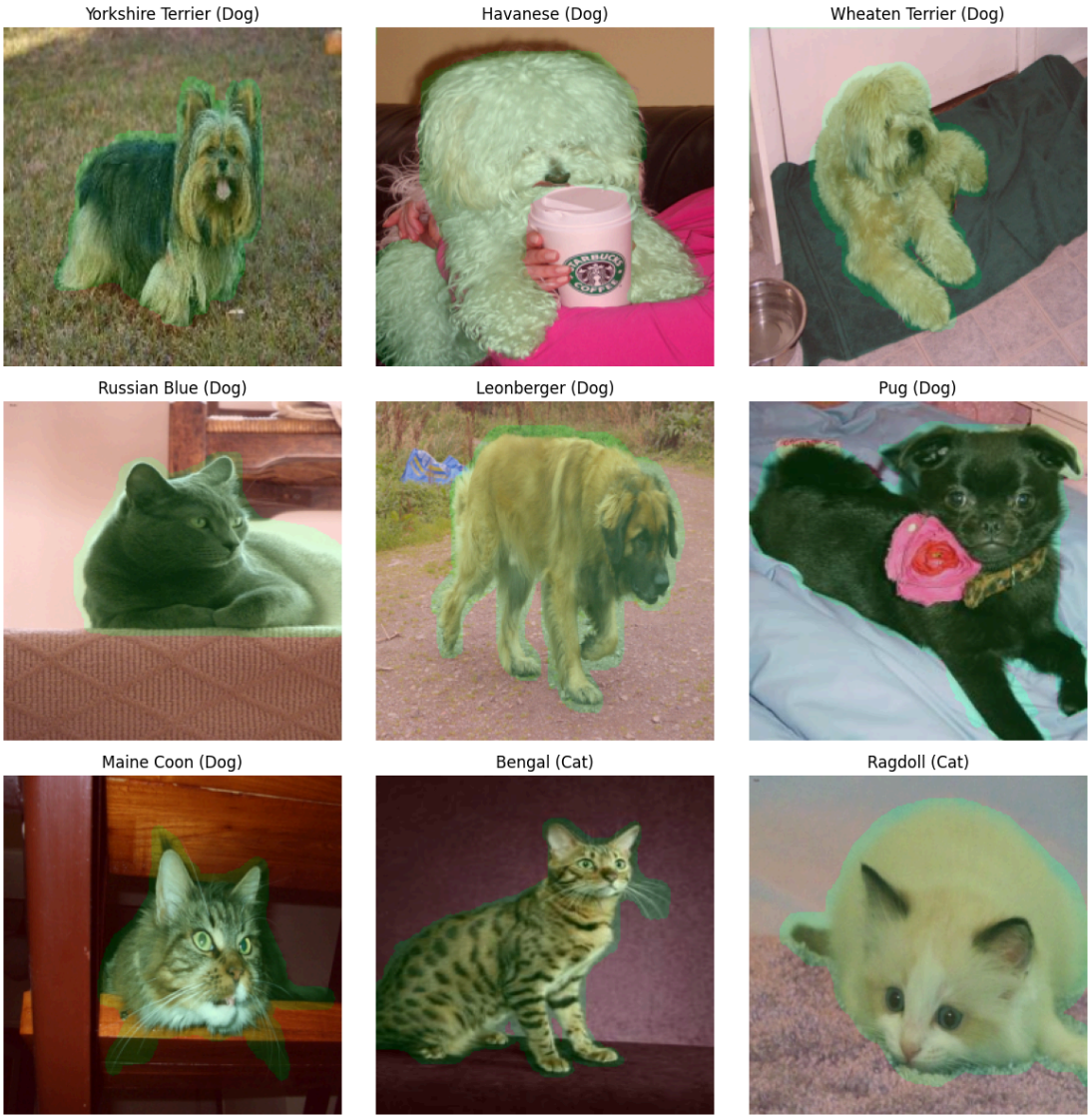
Dataset link - https://www.robots.ox.ac.uk/~vgg/data/pets/

Note - There may be alternate mirrors to download the data. You are allowed to use them.

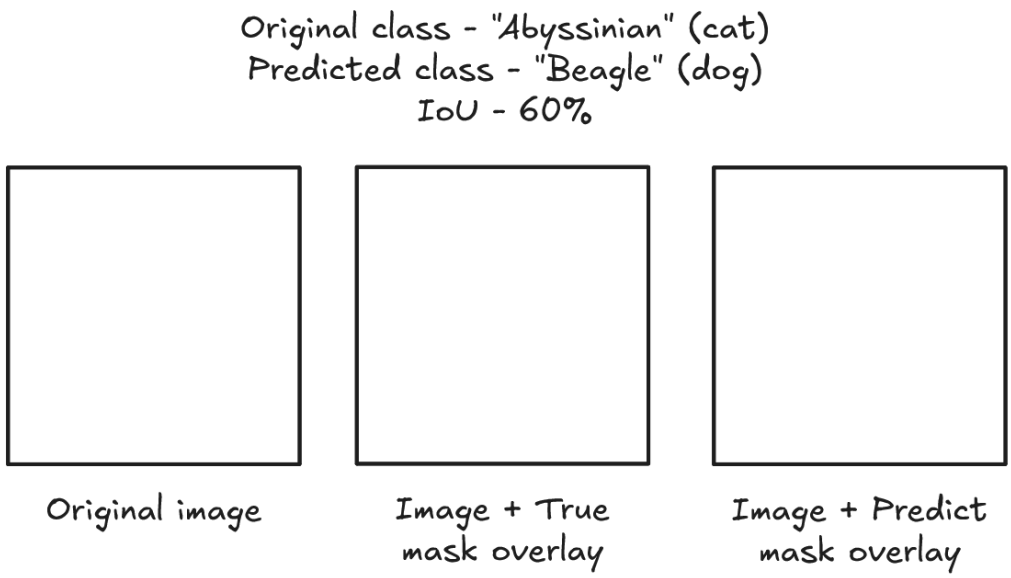
Minimum expectations -

- **Dataset exploration**
  - Load the data and change the unclassified or boundary pixels in the masks to foreground.
  - Prepare a code that takes 9 random indices and plots images with mask overlays in the following 3x3 format -



- **Segmentation (Base U-Net)**
  - Prepare a **U-Net** architecture (https://arxiv.org/abs/1505.04597) to segment the data.

- Attach a classifier head to the encoder output and classify the data. Train together.
- Prepare a code that takes any image index as an input, passes it through your model and displays the results in the form -



For your presentation part, you can adhere to random indices to pass through your model.

- Follow the presentation requirements for the reporting metrics.

- **Segmentation (Attention U-Net)**
  - Update the **U-Net** architecture to **Attention U-Net** (https://arxiv.org/abs/1804.03999) and run the segmentation training again.
  - Attach the previous classifier head to the encoder output and classify the data again. Once again, train together.
  - Present the results in the previous presentation format.
  - Follow the presentation requirements for the reporting metrics.

Open-ended design choices -

- You do not have to separately preprocess each image, you can find a common pattern and apply a preprocessing pipeline. It's your duty to ensure you are getting the best possible results with or without preprocessing.
- The dataset is fairly small considering the number of classes present. You are encouraged to augment the data.
- You can use any classifier architecture. There are no restrictions on which one to use. Whatever you do, try to maximize your accuracy.

Presentation requirements -

Please note that you do not have to make a presentation slide if you opt not to. But in that case, you have to prepare the notebook carefully so that you can present your work through the notebook. Include all the requirements in the notebook. For training, show training statistics (metric update per epoch and loss curves), and result summaries. For the result summaries, we want to see the values of mean Intersection over union (mIoU), dice coefficient and pixel accuracies for the segmentation task, and accuracy, precision, recall and F1 scores for classification task. The result summaries should be presented for all sets - training, validation and test.

PS - If you have prepared the code in VSCode or other IDEs in a modular pattern, you should prepare a presentation slide compiling the results as it may get confusing navigating the project. If you are confident that you can present it in a modular way without getting confused, you can skip the slide, but remember if you cannot present it properly, you lose marks in presentation.

Marks distribution -

1. Attendance (1%)
2. Presentation (2%)
3. Demonstration (6%)
4. Viva (6%)

The attendance and presentation part of the assessment is self-explanatory.

For the demonstration part, we will provide you with some random images on which you have to run your model and get the predictions. You have to present the predictions in the format shown in the minimum expectations. Do note that we may ask you to execute any block of code or simply ask you to explain the function of the code block. Failure to do so will result in mark deduction from this segment under the GPT penalty.

Finally in the viva, we will ask you questions about your understanding of the project. Even if you haven’t worked on one part of the project, you may still be asked from that part. So it's your responsibility to understand the project properly.

Bonus tasks -

Bonus marks will only be awarded at the discretion of the lab faculties. We are providing a list of possible cases where bonus marks may be provided. But remember, if you do not complete the bonus tasks properly, you will not receive any additional marks. Incomplete tasks are worth 0. With that being said, the list of bonus tasks that you may attempt (you can go for multiple bonus tasks) (the tasks are arranged in ascending order of value) -

1. Test out different well-established classifier architectures (like MobileNet, EfficientNet, DenseNet etc.) and see how the performance changes. Show comparison between at least 3. (Difficulty: Easy)

2. Test if your model performance improves when you augment the data. Mention how you have augmented the data and how much the performance improves. (Difficulty: Low medium)

3. Remember how you converted the per pixel decision to a binary problem by including the boundary pixels into foreground? Can you segment without this step? Segment the images keeping the segmentation as a three class problem. Repeat segmentation using both U-Net and Attention U-Net. Use apt metrics. (Difficulty: Medium)

4. Test how your model performance changes with change in hyperparameters. For example, you can test for the most optimal optimizer under a wide range of learning rates. Remember, you have to compile your results properly for your work to be considered complete. (Difficulty: Medium)

5. Implementing EfficientDet (https://arxiv.org/abs/1911.09070) architecture on the decoder side of the original U-Net architecture. (Difficulty: High)

If you want to do anything different for the bonus segment, do let your lab faculties know and get it approved before working on it.